

Everett Richards

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Overview

I am a dedicated computer scientist who is passionate about using technology to make transportation systems more safe, efficient, and equitable. I am currently working towards my PhD in Computer Science at the University of Delaware, working in the CAR Lab with Prof. Weisong Shi.

Education

University of Delaware

PhD Computer Science

Newark, DE

Aug 2026–Present

- Robert W. Gore Fellowship

San Diego State University

B.S. Computer Science • B.S. Applied Mathematics

San Diego, CA

Aug 2023–May 2026

- Most Outstanding Computer Science Graduate
- Most Outstanding Mathematics & Statistics Graduate
- Summa Cum Laude (4.0 GPA)

Research Experience

Doctoral Researcher – University of Delaware

Connected Autonomous Research (CAR) Lab, Prof. Weisong Shi

Aug 2026 – Present

Newark, DE

- Working on state-of-the-art Vehicle-to-Everything (V2X) communication systems.

NSF REU Research Fellow – Worcester Polytechnic Institute

Vision, Intelligence, and System (VIS) Lab, Prof. Ziming Zhang

May–Aug 2025

Worcester, MA

- Conducted robustness evaluations through targeted injection of Gaussian noise into 3D LiDAR point clouds, simulating sensor degradation due to low-resolution sensors and adverse weather conditions.
- Proposed and empirically validated a novel noise-aware training curriculum, achieving up to a 40% improvement in model robustness under challenging real-world scenarios.

NSF REU Research Fellow – UC San Diego

SU Computer Vision Lab, Prof. Hao Su

Sept 2024–June 2025

La Jolla, CA

- Conducted experiments on the impact of Gaussian noise injection in robotic imitation learning models.
- Used MuJoCo, RoboSuite, RoboMimic, and MimicGen for robotic simulation and imitation learning.

NSF REU Research Fellow – University of Delaware

Intelligent Edge Systems (IES) Lab, Prof. Lena Mashayekhy

June–Aug 2024

Newark, DE

- Developed and validated two algorithms to improve object detection accuracy in autonomous vehicles.
- Achieved up to 70% accuracy improvement over legacy methods, validated on a four-robot testbed.

Publications

- Y. Wang, **E. Richards**, W. Shi. “Data Infrastructure for Robotaxi Services: Opportunities and Challenges.” IEEE Internet Computing Magazine, Volume 30, Issue 5. 2026.
- **E. Richards**. “Task-Aligned Stability Analysis of Vision-Language Models for Autonomous Driving Hazard Detection.” 2026 International Conference on Machine Learning (ICML) Workshop on Combining Theory and Benchmarks. Seoul, South Korea, 2026.
- **E. Richards** and L. Dai. “Modeling Imitation Learning Robustness to Noisy Demonstrations via Sigmoid Degradation.” 2025 IEEE MIT Undergraduate Research Technology Conference (URTC). Cambridge, MA, 2025.
- **E. Richards**, A. Lopez, J. Morales, and Z. Zhang. “From Chaos to Clarify: Strengthening 3D Collaborative Autonomous Vehicle Perception with Noise-Aware Training.” 2025 IEEE MIT Undergraduate Research Technology Conference (URTC). Cambridge, MA, 2025.
- **E. Richards**, B. Thapa, and L. Mashayekhy. “Edge-Enabled Collaborative Object Detection for Real-Time Multi-Vehicle Perception.” 2025 IEEE International Conference on Edge Computing and Communications (EDGE). Helsinki, Finland, 2025, pp. 13-22, doi: [10.1109/EDGE67623.2025.00011](https://doi.org/10.1109/EDGE67623.2025.00011).

Instructional Experience

Teaching Assistant, Computer Science Aug 2026–Present
University of Delaware Newark, DE

- Teaching Assistant for the Department of Computer and Information Sciences.

Coding Camp Instructor June–Aug 2026
iD Tech Camps / UC San Diego La Jolla, CA

- Instructed students aged 10-12 on computer programming (Lua, Python) and video game development.

Tutor, Mathematics & Computer Science Aug 2024–May 2025
SDSU Math and Science Learning Center (MSLC) San Diego, CA

- Tutored students in computer science (Java, Python, data structures, algorithms), mathematics (calculus, discrete math, real analysis), and physics.
- Participated in professional development workshops to enhance tutoring skills.
- Certified Level 1 Tutor by the College Reading and Learning Association (CRLA).

Instructional Assistant, Discrete Mathematics Jan 2024–May 2025
SDSU Dept. of Mathematics and Statistics – Instructor: Vadim Ponomarenko San Diego, CA

- Led weekly office hours for ten groups of 6 to 8 students, covering proof techniques, recursion, and boolean algebra.
- Developed and discussed weekly focus exercises to reinforce key topics.

Leadership Experience

Senator *pro tempore* Sept 2026–Present
UD Graduate Student Government ([link](#)) Newark, DE

- Represented hundreds of graduate students in the UD Department of Computer and Information Sciences at GSG meetings.

Student Advisory Council Member Sept 2026–Present
UD Biden Institute ([link](#)) Newark, DE

- Served on a council that oversees the Biden Institute, a nonpartisan policy hub affiliated with the Joseph R. Biden, Jr. School of Public Policy at UD.

Hackathon Founder & Co-Director Jan–Nov 2025
Innovate 4 SDSU Hackathon ([link](#)) San Diego, CA

- Led two separate weekend-long on-campus hackathons where more than 200 students competed to develop projects aimed at improving student life.
- Secured more than \$10,000 in funds from industry partners including Google, Qualcomm, and Sony.
- Supervised and delegated tasks to a team of 30 volunteers and planning committee members.

President, ACM SDSU Apr 2024–May 2026
Association for Computing Machinery (ACM) Student Chapter at SDSU ([link](#)) San Diego, CA

- Founded and led SDSU's chapter of the world's largest professional organization dedicated to computing.
- Secured a \$2,500 sponsorship from Google to fund events and initiatives.
- Organized and personally presented dozens of technical workshops on machine learning, web development, and version control, and other topics.

Vice President, CTRL SDSU May 2025–May 2026
Coalition of Tech Representatives and Leadership (CTRL) at SDSU ([link](#)) San Diego, CA

- Served on a board of leaders from various Computer Science student organizations at SDSU.
- Represented CTRL at department meetings and the Industry Advisory Board (IAB).
- Oversaw the NXP Engineering Bootcamp, in which 15 students worked with NXP industry mentors over the course of the semester to create AI-driven solutions to chip design problems.

University Council Representative May 2025–May 2026
Associated Students of San Diego State University San Diego, CA

- Represent 8,000 College of Sciences students on the University Council.
- Oversee financial appropriations for a non-profit with over \$45,000,000 in annual revenue.
- Drafted a successful [resolution](#) advocating for increased transparency in changes to mandatory student fees.

Volunteering at Conferences and Competitions

Conference Support Volunteer , IEEE Conference on Communications and Network Security	Sept 2026
Hackathon Mentor, Panelist, & Judge , <i>CipherHacks San Diego</i> High School Hackathon	June 2026
Remote Competition Judge , National Collegiate Cyber Defense Competition	Feb 2026
Hackathon Founder & Co-Director , <i>Innovate 4 SDSU</i> Collegiate Hackathon	Apr 2025 & Nov 2025
Integration Bee Planning Committee , SDSU Society of Physics Students	Dec 2024
High-Impact Project (HIP) Panelist , SDSU Weber Honors College	Sept 2024 & Sept 2025

Awards and Honors

Academic Awards:

- Most Outstanding Computer Science Graduate, SDSU Dept. of Computer Science (2026)
- Most Outstanding Mathematics Graduate, SDSU Dept. of Mathematics and Statistics (2026)
- Cert. of Excellence in Advanced Programming Languages, World Computing Organization (2025)
- Dean's List, SDSU College of Sciences (2023-2026)
- Social Science Department Award, Poway High School (2023)

Scholarships & Fellowships:

- Robert W. Gore Engineering Fellowship (2026)
- UPE ACM Scholarship (2025)
- Steven Michael Rogers Scholarship in Mathematics (2025)
- George A. Hansen Scholarship (2025)
- Deloitte Foundation Scholarship (2024)
- Mensa Foundation Scholarship (2024)
- Raytheon Systems Scholarship (2024)
- Intuit STEM Scholarship (2023)

Skills and Strengths

Technical Skills:

- Multi-agent autonomous vehicle simulation with ROS2 and CARLA
- Implementing deep learning networks with PyTorch, OpenCV
- Designing parallelized algorithms that perform well across various GPU and CPU configurations
- Python programming and data analysis with NumPy, CuPy, SymPy, Pandas, Matplotlib, and Scikit-learn
- Systems-level programming with C/C++
- Version control and collaborative coding using Git/GitHub
- Multi-modal computer vision, including CNN and ViT, using 2D (camera) and 3D (LiDAR) sensors
- Computer networking and edge computing
- Mathematical data science (compression, information theory, neural networks) and theoretical machine learning

Professional Skills:

- Research, literature review, and technical writing
- Public speaking and presentations
- Leadership and organizational management
- Collaborating with peers with diverse skill sets and backgrounds to achieve common goals
- Tutoring and mentoring students in various age groups (high school and university)
- Drafting L^AT_EX documents
- Publishing peer-reviewed research papers at reputable conferences (ICML, IEEE EDGE, etc.)